

DESIGN AND OPTIMIZATION OF BIOPROCESSES BY SIMULATORS

METHANOL PRODUCTION FROM BIOGAS

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1. Theoretical Introduction

The transformation of biogas, which is predominantly made up of methane (CH_4) and carbon dioxide (CO_2), into methanol represents an environmentally friendly substitute for the conventional method of synthesis from fossil fuels (Bühlmann et al., 2023). This technology employs carbon-free sources to synthesize "green" methanol, which serves as a fundamental raw material in the chemical sector and energy storage systems (Del Pópolo Grzona et al., 2024).

The production of syngas constitutes the first step of the process. Although some technologies necessitate the purification of biomethane, reforming raw biogas (60% CH_4 and 40% CO_2) can be an effective and economically efficient option (Del Pópolo Grzona et al., 2024). In order to initiate the endothermic reaction, the feedstock needs to be heated to 410.7°C . The rate kinetics of steam reforming and the WGS reaction are formulated according to the original equations of Xu and Froment (1989).

The reaction for producing methanol is exothermic and limited by equilibrium reactions and is highly favored by elevated pressure levels (Del Pópolo Grzona et al., 2024). The reaction exhibits poor per-pass conversions, hence, incorporating a substantial recycle stream to optimize the carbon utilization ratio. Though elaborate kinetic models are available, steady-state modeling is performed using Equilibrium Reactors with manipulated outlet compositions to ensure numerical stability and industrial yields (Polierer et al., 2020).

The 1st step entails purification of crude methanol through distillation. Methanol being a polar compound, the Non-Random Two-Liquid (NRTL) approach is applied to attain thermodynamic precision (Cahyaning Ananda et al., 2024). Besides, thermal integration is accomplished by extracting heat from the exothermic methanol synthesis reaction to preheat the distillation column reboiler, ensuring self-reliant operation within the plant (Del Pópolo Grzona et al., 2024).

2. Objectives

This project aims to simulate a biogas-to-methanol plant using UniSim, converting raw biogas (60% , 40%) into 99.5% pure methanol. Objectives include implementing Xu-Froment kinetics, optimizing a high-pressure synthesis loop with heat integration, and ensuring plant thermal self-sufficiency.

3. Calculation Basis

The simulation was performed using two simulators: Honeywell UniSim Design R510 and ProSim 3 Plus. The calculation basis is established to ensure a realistic industrial scale consistent mass balances, and thermodynamic accuracy.

3.1. PLANT CAPACITY AND OPERATION

Annual Production: Based on a product flow of 9.86 kgmole/h of methanol (approx. 316 kg/h), the plant produces roughly 2528 metric tonnes per year.

Final flows differ slightly between simulations as discussed in Table 2 however, UniSim results were used as the primary basis for this calculation.

Operating Hours: 8,000 hours/year (approx. 91% availability due to possible maintenance).

Target Purity: 99.5% mass purity of Methanol.

3.2. THERMODYNAMIC PACKAGE

For the reforming process, the Peng–Robinson was chosen because of its high suitability in modeling reactions in the gas phase at higher temperatures, especially when the system includes hydrocarbons and syngas.

The NRTL model was used for the synthesis process and the purification stage. The model is ideal for vapor-liquid equilibrium calculations in polar systems like that of methanol and water, and it is also able to estimate the gas solubility at high pressures.

3.3. REACTOR AND EQUIPMENT SPECIFICATIONS

The reformer reactor was represented as a PFR (Plug Flow reactor) in UniSim, using the kinetics that were derived by Xu and Froment in 1988. A Gibbs reactor based on the minimization of Gibbs free energy was implemented in ProSim to simulate the methanol production process from biogas. Initially, a kinetic reactor model equivalent to the one

developed in UniSim Design was tested under the same operating, geometry and kinetic conditions,. However, despite using identical inputs, the kinetic model did not converge properly in ProSimPlus due to differences in the internal numerical implementation of heterogeneous reaction kinetics, phase equilibrium calculations, pressure partial handling, and solver robustness between both simulators

The reactor for the production of methanol was designed as an equilibrium reactor that would operate at 6000 kPa. In order to simulate actual conditions better, which could not be achieved due to kinetic convergence problems in UniSim and ProSim, the output streams from the reactor were mixed to get a 99.5% conversion of reactants.

A large amount of the recycle stream of 253.5 kmol/hr was considered in the design in UniSim, corresponding to a recycle to fresh feed ratio of about 15:1. The recycle stream corresponds to 135.2 kmol/hr in ProSim, the recycle to fresh feed ratio is around 17:1.

3.4. ENERGY INTEGRATION BASIS

The energy integration basis focuses on internal heat recovery to minimize external utility costs. With a total cooling load of 1,620.5 kW (primarily from E-101 and E-104) and a heating demand of 1,011.4 kW, for Unisim and 1608 kW load and 1338 kW demand for Prosim, as discussed in Table 1. In both cases, the process is a net heat producer. A thorough comparison can be seen in Table 1.

Integration is achieved by using high-grade waste heat from the reformer and reactor to drive the Distillation Reboiler and generate process steam. Additionally, cooling syngas minimizes compression work, ensuring high thermal efficiency and reduced OPEX.

4. Flowchart and its Description

The process flowsheet for the production of green methanol from biogas through all steps of the process to achieve purity in the final methanol product as proposed by Honeywell UniSim Design (R510) is shown in Figure 1, and by ProSim3 Plus is presented in Figure 2. This shows how the three key functional processes work together in the complete process flowsheet.

4.1. REACTOR MODELING ADJUSTMENTS

In UniSim, reforming utilizes Xu and Froment (1989) kinetics for accuracy. Conversely, methanol synthesis was modeled as an Equilibrium Reactor with blended outlets to ensure numerical stability and mimic industrial performance, bypassing convergence issues found in multitubular kinetic models. This adjustment maintains alignment with literature results while ensuring reliable steady-state convergence. In ProSim, both reforming and methanol synthesis were modeled through Gibbs Reactor, producing equilibrium compositions and overall process results comparable to those obtained in UniSim, it was considered an appropriate and reliable alternative for the process simulation of the biomethanol production.

4.2. PROCESS DESCRIPTION

A. Feed Conditioning and Reforming

The “Feed (Biogas)” enters the facility at 25°C and 1000 kPa with a flow rate of 16.74 kgmole/hr and contains 60% CH₄ and 40% CO₂ on a mole basis. The feedstock is preheated to 410.7°C. It is combined with high-pressure steam at 550°C and reacted in a reformer. At this point, the biogas undergoes steam reforming, producing synthesis gas that is a combination of H₂, CO, and CO₂.

B. Methanol Synthesis Loop

Syngas leaving the process is cooled for the removal of extra moisture content and then subjected to compression (K-101) to increase pressure to 6000 kPa. After being compressed, the gas mixture is blended with high recycle flow rate, with adjustment to the reactor inlet temperature of 220 °C. The reaction between hydrogen and carbon oxides takes place at an equilibrium reactor to produce methanol.

C. Separation, Purification, and Recycling

The product exiting the methanol reactor is cooled and subjected to multi-stage flash separation. The flash separation process is carried out in Flash Drum 2 where liquid crude methanol and non-reacted gaseous components are separated at 6000 kPa pressure. The latter component is then recycled back into the reactor to alleviate equilibrium problems. Further separation is carried out in Flash Drum 3 at 100 kPa to remove any gas dissolved in the liquid phase prior to feeding it into the distillation column (COL1).

The distillation column specifications implemented in ProSimPlus differ from those used in UniSim Design due to differences in the specification structure and operational constraints available in each simulator. In ProSim, the selection and combination of column specifications and adjusted variables are more restrictive, particularly for columns with partial condensers. As a result, alternative specification combinations were required to achieve numerical convergence and stable operation while maintaining process behavior and separation performance comparable to those obtained in UniSim.

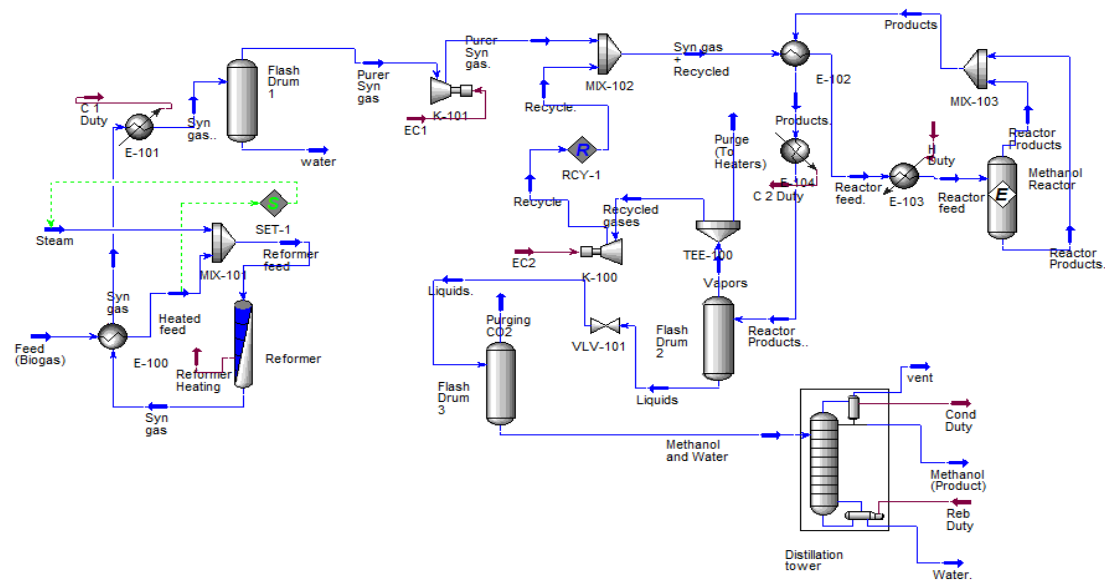


Figure 1. Flow Chart of the process in UniSim.

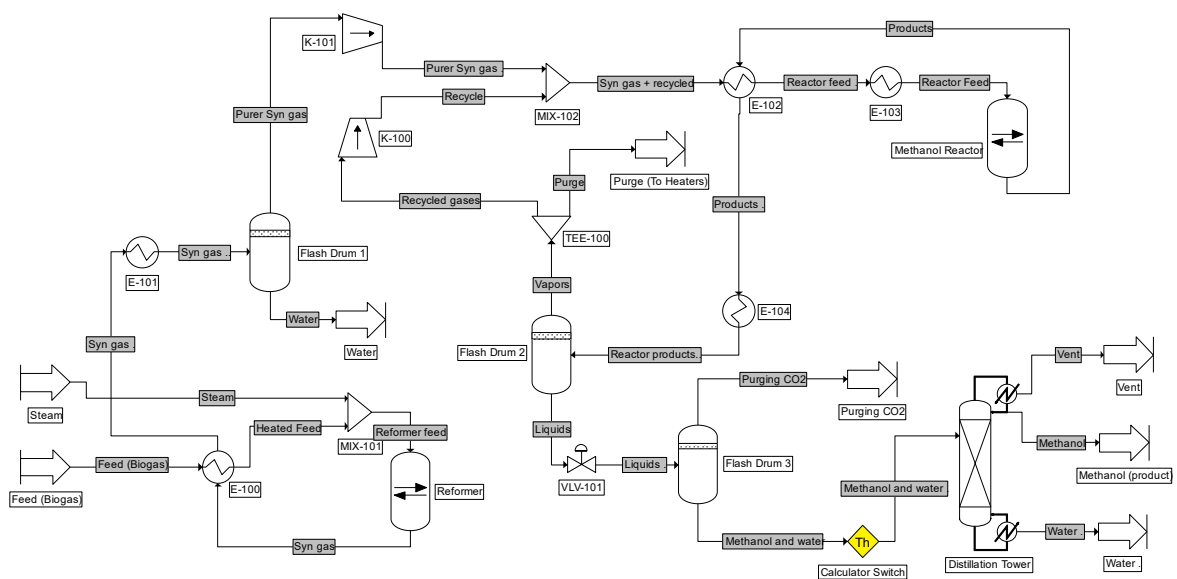


Figure 2. Flow Chart of the process in UniSim.

5. Results and Discussion

The simulation has been successful in representing the conversion of crude biogas to pure methanol by means of an altered technique according to the specifications provided by Izurieta et al., (2018). This process does not incur any costs for the initial use of membranes since the process takes place using 60% of methane and 40% of carbon dioxide. The analysis presented below is conducted based on the energy intensity and mass balance of the process.

5.1. ENERGY NEEDS AND HEAT INTEGRATION

The energy needs of the plant depend on the temperature levels in the reformer stage followed by cooling needed for the syngas treatment in the synthesis process. The following table shows some energy duties determined for major equipment used in the process.

Table 1. Major Equipment Energy Duties

Feature	ProSim Simulation	UniSim Simulation
Total Heating Demand	1,388 kW	1,011 kW
Total Cooling Load	1,608 kW	1,620 kW
Specific Energy Intensity	184.5 Mcal/kmol	102.5 Mcal/kmol

Based on the energy evaluation, a large difference is noted between the two simulators due to heating demand and Specific Energy Intensity (SEI). The total cooling load is found to be almost identical in both models (less than 1%) but there is an estimated increase of about 27% in the total heating demand of ProSim relative to UniSim.

This translates to a considerable difference in the specific energy intensity, which was found to be lower in UniSim at 102.5 Mcal/kmol versus ProSim's figure of 184.5 Mcal/kmol. This shows a difference of 44%. The Prosim simulation requires more energy due to equilibrium-based (Gibbs) reactors we had to use and more restrictive column specifications.

Integration is achieved by using high-grade waste heat from the reformer and reactor to drive the Distillation Reboiler and generate process steam. Additionally, cooling syngas minimizes compression work, ensuring high thermal efficiency and reduced OPEX.

5.2. KEY STREAM RESULTS

The mass balance calculation as simulated in Unisim through the main stages of the plant indicates high efficiency in the use of carbon. Table 2 shows the temperatures, pressures, and molar flows for the feed input, reactor feed (optimized), high recycle stream, and output products.

Table 2. Key stream results

Stream Name	Property	UniSim Design	ProSim Plus
Feed (Biogas)	Temperature (°C)	25.0	25.0
	Pressure (kPa)	1,000	1,000
	Molar Flow (kgmole/h)	16.74	16.74
	Composition	CH ₄ (0.60), CO ₂ (0.40)	CH ₄ (0.60), CO ₂ (0.40)
Recycle	Temperature (°C)	25.52	24.85
	Pressure (kPa)	6,000	6,000
	Molar Flow (kgmole/h)	253.5	285.2
Methanol (Product)	Molar Flow (kgmole/h)	9.86	7.51
	Methanol Fraction	0.995	0.998

The differences between the two cases are negligible, and this can be attributed to the types of reactors chosen for simulation. The UniSim simulator used an Equilibrium Reactor, while in ProSim we used a Gibbs reactor because of the constraints of the program.

Unisim simulation was able to produce higher amounts of methanol at 9.86 kgmole/h. On the other hand, the ProSim simulation was able to produce lesser methanol output (7.51 kgmole/h) but yielded greater purity (0.998).

The slight difference in the "per-pass" conversion accounts for the recycle loop changes, which saw ProSim using a higher recycle flow of 285.2 kgmole/h versus UniSim's recycle flow of 253.5 kgmole/h. Basically, Gibbs Reactor gives conservative estimates of the extent of

reactions, while Equilibrium Reactor assumes stoichiometric reactions. Both programs are still consistent with each other, from a mathematical point of view.

5.3. DISCUSSION OF THE RESULTS

From the results of the simulations, it is clear that the direct conversion of raw biogas (containing 60% CH₄ and 40% CO₂) to produce methanol is feasible.

It is obvious that the very high heat duty of the heaters results from the highly endothermic reforming reactions. The most important aspect regarding this part of the process is the operation of cooler E-101. This cooling effect can be considered as the main conditioner for the syngas before further compression since cooling down reduces the effect of water, which is produced, acting as a diluent.

One of the important observations from the simulation stage was the numerical instability associated with Van den Bussche and Froment (1996) kinetic models within the UniSim software. Despite the kinetically correct formulation, this equation can easily lead to simulation divergence within complicated recycle systems. The use of an Equilibrium Reactor with "blended outlets" turned out to be an efficient engineering solution. In this way, it was possible to obtain stable steady-state operation while ensuring the correct product composition, corresponding to the experimentally established results by Izurieta et al. (2018). The reason for this engineering approach lies in the kinetic behavior of industrial reactors, which cannot achieve theoretical equilibrium completely.

Low per-pass conversion in methanol synthesis requires such a large recycle stream of 253.5 kgmol/h for Unisim and 285 kmol/h for Prosim. The size of the recycle loop is close to 15 times greater than the feed gas stream for Unisim and 17 times for Prosim. Without such a big recycle stream of H₂ and CO_x gases, the process becomes economically inefficient because of very bad carbon efficiency.

Energy considerations in the plant provide an example of successful process intensification. By capturing the heat of reaction in the methanol reactor to produce high-pressure steam, the system meets the reboiler load of required by the distillation column. This internal heat

exchange plays a vital role in the "green" production of methanol, which reduces dependence on external utilities that use fossil fuels for producing pure methanol.

6. Conclusions

The study proves that there is technically the possibility of manufacturing highly pure green methanol from biogas having a methane content of 60%, while the remaining 40% consists of carbon dioxide. Through direct use of raw biogas and bypassing of its transformation into biomethane, this method can eliminate the cost implications of energy consumption during membrane separation.

For comparison, UniSim provides superior energy efficiency and throughput (9.86 kmol/h) by using kinetic modeling. ProSim achieves higher purity (99.8%) but requires more energy due to equilibrium-based (Gibbs) reactors we had to use and more restrictive column specifications.

The process ensures conformity to product specifications, obtaining methanol of 99.5% purity by mass, equivalent to industry "AA" grades. The degree of methanol purification was accomplished through a 30-stage distillation column using the NRTL-Ideal thermodynamics package.

The operational performance was improved by employing a higher recycle ratio of 15:1 in case of Unisim and 17:1 for Prosim to compensate for the thermodynamic constraints associated with the formation of methanol.

The process exhibits impressive energy integration. The thermal self sufficiency of the separation section is attained by using the exothermic heat generated during the methanol formation to produce steam, which fully covers the heat duty of the reboiler for both cases.

The above-mentioned simulation technique is highly effective in validating the flowsheet, which shows that green methanol synthesis will still be sustainable even if the mathematical modeling framework or reactor-specific algorithm differs from what was used in the design phase.

7. Recommendations

In order to improve the design process and assess its industrial feasibility, the following suggestions are recommended:

A complete cost analysis must be carried out to estimate the capital and operational costs. In addition, the balance between removing biogas upgrading and larger equipment to accommodate the extra carbon dioxide must be considered.

Further research will involve moving from the current steady-state modeling to dynamic modeling. This will help assess the flexibility of the process under varying input streams, which is often expected in practical anaerobic digestion processes.

Moreover, other heat recovery possibilities may be considered. For instance, the potential use of the purge stream as fuel for other process heaters and the reformer furnace would help reduce the dependency on energy from external sources.

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